Do Now · retrieval

Lesson 083 · Dijkstra: Shortest Paths with Non-negative Weights

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does relaxation do?

Q2

What is a weighted edge?

Q3

What condition does standard Dijkstra require?

Q4

Undirected graph edges A–B, A–C, B–D. Give BFS order from A visiting B before C.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

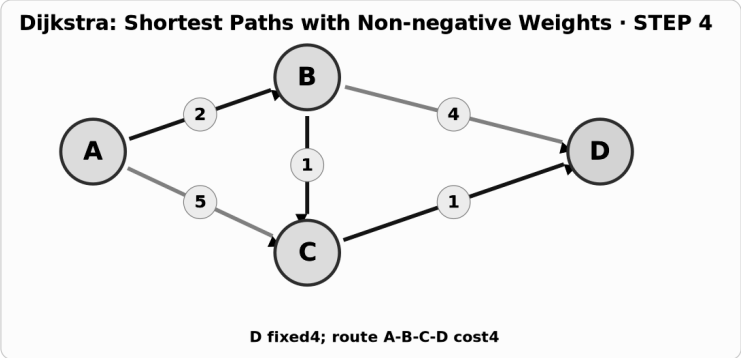
Core idea 2

Apply — Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

Core idea 3

Evaluate — Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Dijkstra — Non-negative weighted shortest paths.

tentative distance — Current best-known route cost.

relaxation — Replace distance if improved.

predecessor — Previous vertex on best path.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Find the shortest P→R route. Directed edges: P→Q = 3, P→R = 8, Q→R = 2.

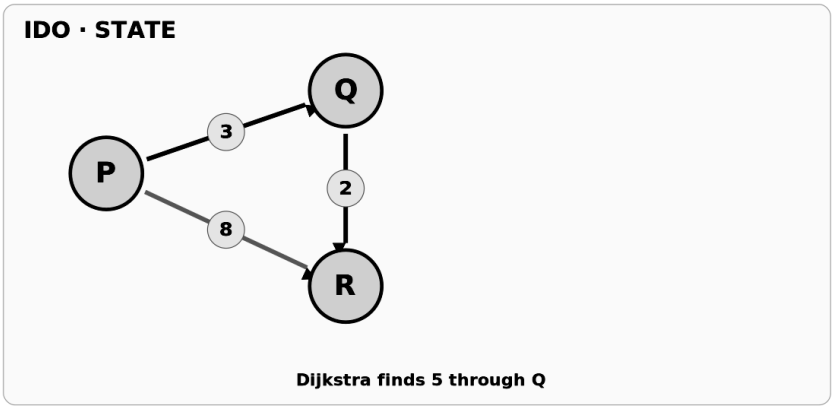
Follow the worked method

1. Update neighbours — Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.

2. Improve the tentative route — Q=3 is the smallest unsettled distance. Via Q→R, the candidate is 3+2=5 . This beats R=8, so update R=5 and predecessor R←Q. Settled now: P,Q.

3. Reconstruct the actual path — R=5 is next, and is the target. Follow predecessor pointers R←Q←P ; reverse to obtain P→Q→R , total cost 5 . No vertex with a larger tentative distance was prematurely finalised.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Settle X; calculate both tentative distances
Start at X=0. Which neighbour gets distance 4? Which gets 10? Which unsettled vertex should Dijkstra select next?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

From A in stated graph, give tentative distances after settling A and B. [6]

Question 2

Continue from C: give final shortest cost and predecessor chain for D. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why negative edge weights can invalidate Dijkstra's finalisation rule, using a counterexample. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Dijkstra is guaranteed correct on graphs with arbitrary negative edge costs.

Use key vocabulary from memory

1. Explain Dijkstra in your own words.

2. Explain tentative distance in your own words.

3. Explain relaxation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.